Empirical Scaling Analysis of Representation Norms in Vision Models

1. Introduction

Scaling laws describe how the behavior of machine learning models changes as a function of model size, dataset size, and compute. While such laws have been extensively studied in natural language processing, their empirical behavior in computer vision remains comparatively less explored, especially at the level of internal representations. Understanding how vision models scale can provide insight into architectural design choices and guide the development of more efficient models.

This internship project focuses on an empirical investigation of scaling behavior in convolutional neural networks for computer vision. Instead of studying task-level performance metrics such as accuracy or loss, the study analyzes how **internal feature representations** scale with increasing model capacity and dataset size. This approach enables controlled analysis without relying on labeled supervision and avoids confounding effects introduced by task-specific objectives.

The work explores two primary axes of scaling:

1. **Model size scaling**, by comparing variants within and across architecture families.
2. **Dataset size scaling**, by analyzing representation behavior under increasing data availability.

2. Background and Prior Foundations

Empirical scaling behavior in neural networks has been most prominently studied in the context of large-scale language models, where predictable relationships between model size, dataset size, and optimization loss have been observed. These studies demonstrated that increasing scale often leads to smooth and systematic improvements in model behavior, motivating further investigation into scaling phenomena across different domains.

In computer vision, convolutional neural networks remain the dominant architectural paradigm. Residual Networks (ResNet) introduced skip connections to enable the stable training of deeper models, while DenseNet proposed dense connectivity patterns to encourage feature reuse and improve parameter efficiency. Both architectures have been extensively validated on large-scale image classification benchmarks such as ImageNet and represent distinct design philosophies for scaling model capacity.

Most prior vision research evaluates scaling primarily through supervised performance metrics such as accuracy or loss. However, these metrics conflate representation quality with task-specific objectives and labeled supervision. As a result, they provide limited insight into how **internal representations themselves evolve** with increasing model or dataset size.

This project builds upon these foundational ideas by focusing on **representation-level analysis** rather than task performance. By examining how feature norms scale across architectures and data regimes under controlled conditions, the study offers an empirical perspective on scaling behavior in vision models that complements existing performance-driven evaluations.

3. Dataset Description

All experiments were conducted using a subset of the ImageNet dataset. To reduce class imbalance and simplify interpretation, a **single-class subset (tench)** was used. This choice allows the analysis to focus on representation behavior without introducing additional variability due to multi-class decision boundaries.

Two experimental setups were defined:

- A fixed dataset of 200 images for model-size ablation.
- Three datasets of sizes 100, 200, and 300 images for dataset-size ablation.

All images were resized to a fixed resolution and processed using identical preprocessing pipelines across experiments.

4. Methodology

4.1 Representation Metric

Since the study operates on unlabeled data, conventional supervised metrics such as classification loss or accuracy are not meaningful. Instead, the **ℓ2 norm of final-layer feature representations** was used as the primary metric.

For each input image, the feature vector produced by the final projection layer was extracted, and its ℓ2 norm was computed. This metric serves as a proxy for representation scale and allows comparison across architectures and data regimes.

All backbone parameters were frozen, and only the final projection layer was trained using a deterministic protocol. This ensures that observed trends arise from architectural and data differences rather than stochastic training effects.

4.2 Model Size Ablation

To study the effect of model capacity, two architecture families were considered:

- **ResNet family**: ResNet-18, ResNet-50, ResNet-101
- **DenseNet family**: DenseNet-121, DenseNet-169, DenseNet-201

All models were initialized with pretrained weights and evaluated on the same fixed dataset of 200 images. The experimental setup was identical across models, enabling fair within-family and cross-family comparisons.

The goal of this analysis was to examine how representation norms scale as model depth and capacity increase within each architecture family.

4.3 Dataset Size Ablation

To isolate the effect of dataset size, a fixed architecture was used while varying the number of training samples. Experiments were conducted using:

- ResNet-18
- DenseNet-121

For each architecture, datasets of size 100, 200, and 300 images were used. All other factors, including preprocessing, training protocol, and metric computation, were held constant.

This setup enables analysis of how representation norms evolve with increasing data availability.

4.4 Reproducibility and Experimental Control

All experiments were conducted under a strictly controlled and reproducible setup:

- Fixed random seeds
- Deterministic backend settings
- Frozen backbones
- Identical data ordering and preprocessing

These measures ensure that observed trends reflect systematic scaling behavior rather than stochastic variation.

5. Analysis and Results

5.1 Model Size Scaling Results

The model-size ablation revealed consistent trends within architecture families. Mean feature norms exhibited systematic variation as model capacity increased, indicating that deeper models produce representations with different scaling characteristics.

Comparing ResNet and DenseNet families showed distinct scaling behaviors, suggesting that architectural connectivity patterns influence how representations evolve with increased capacity. While both families demonstrated structured trends, the magnitude and smoothness of scaling differed across architectures.

5.2 Dataset Size Scaling Results

The dataset-size ablation showed a weaker but observable dependence of representation norms on dataset size. Increasing the number of samples led to changes in the average feature norm, although the effect was less pronounced than that observed for model size.

Comparisons between ResNet-18 and DenseNet-121 indicated that architectural differences also influence sensitivity to dataset size, reinforcing the role of model design in representation scaling.

6. Empirical Observations

1. **Systematic scaling of representation norms with model capacity**
   Across both architecture families, increasing model depth and capacity leads to consistent changes in the magnitude of feature representations, indicating a structured relationship between model size and internal representation scale.

2. **Architecture-dependent scaling behavior**
   ResNet and DenseNet models exhibit distinct scaling patterns, suggesting that architectural design choices—such as residual versus dense connectivity—influence how representations evolve as model capacity increases.

3. **Stronger sensitivity to model size than dataset size**
   Within the examined range, variations in model capacity produce more pronounced changes in representation norms than comparable increases in dataset size, indicating that architectural scaling has a larger impact on representation behavior under the studied conditions.

Analysis of Feature Norm Scaling Across ResNet and DenseNet Architectures

1. Introduction

This section presents a detailed quantitative analysis of feature norm values obtained from multiple convolutional neural network architectures spanning two families: ResNet (ResNet18, ResNet50, ResNet101) and DenseNet (DenseNet121, DenseNet169, DenseNet201), evaluated on unlabeled image data. The objective is to examine how model depth and architectural design influence feature magnitude, identify intra-family scaling patterns, and analyze differences in scaling behavior between residual and densely connected networks.

2. Descriptive Statistics

Descriptive statistics were computed for all DenseNet and ResNet models to summarize the distribution of feature norm values. Key statistics include mean, median, standard deviation, minimum, maximum, and quartiles. **DenseNet Family – Descriptive Statistics**

Model	Mean Feature Norm	Median	Std Dev	Min	25%	75%	Max
DenseNet121	14.7422	14.2295	2.3586	10.1229	13.3426	15.4963	26.3886
DenseNet169	17.2423	16.8160	2.7863	12.2849	15.1691	18.7678	28.5573
DenseNet201	14.0086	13.7197	2.4182	8.3496	12.3604	15.2233	23.2330

Observation:

- DenseNet169 exhibits the highest mean and median feature norms within the DenseNet family, indicating stronger overall feature activation with increased depth.
- DenseNet121 and DenseNet201 show comparable distributions, suggesting diminishing returns in feature magnitude beyond a certain depth.
- The relatively low standard deviation across all DenseNet variants indicates stable and consistent feature representations.

ResNet Family – Descriptive Statistics

Descriptive statistics were computed for all ResNet models to summarize the distribution of feature norm values.

Model	Mean Feature Norm	Median	Std Dev	Min	25%	75%	Max
ResNet18	18.1519	17.8439	2.6730	12.9380	16.0735	19.8330	25.2001
ResNet50	7.0626	3.7367	26.6715	2.3576	3.1234	4.4682	297.2228
ResNet101	475.5933	4.7335	2504.8860	2.4694	3.7681	13.1498	23656.3672

Observation:

- ResNet18 shows stable feature norms with low variance, indicating controlled feature scaling in shallow architectures.
- ResNet50 and ResNet101 exhibit extreme variance and heavy-tailed distributions, driven by a small number of samples with very large feature norms.
- The large gap between median and mean for deeper ResNet models indicates strong outlier sensitivity as depth increases.

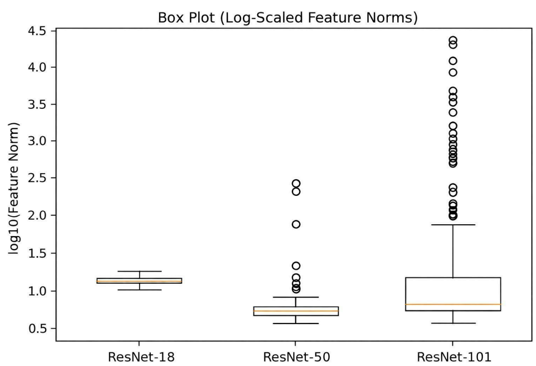
Comparison of Descriptive Statistics: ResNet vs DenseNet

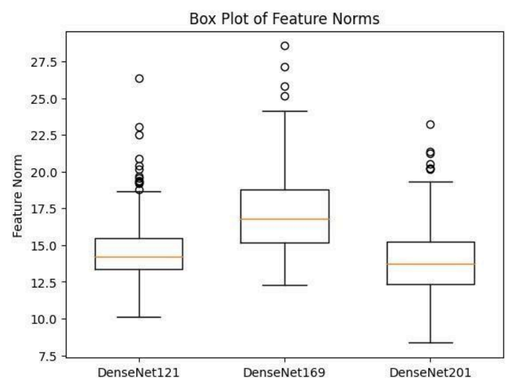
- **Central Tendency:**
  DenseNet models show tightly clustered mean feature norms in the range 14.0–17.2, with medians closely matching the means. ResNet models show inconsistent central tendency, with ResNet18 having a mean of 18.15, ResNet50 7.06, and ResNet101 475.59, while their medians remain low (3.7–4.7 for deeper models).

- **Dispersion:**
  DenseNet models exhibit low and stable standard deviations (2.36–2.79) across depths. In contrast, ResNet models show rapidly increasing dispersion, with standard deviations of 2.67 (ResNet18), 26.67 (ResNet50), and 2504.89 (ResNet101).

- **Range and Extremes:**
  DenseNet maximum feature norms remain below 30 across all variants. ResNet architectures show significantly larger ranges, with maximum values of 297.22 for ResNet50 and 23,656.37 for ResNet101.

- **Quartile Spread:**
  The interquartile ranges for DenseNet models remain narrow and consistent, while ResNet50 and ResNet101 show large gaps between quartiles and extreme maxima, indicating skewed distributions.





3. Correlation Analysis

A Pearson correlation matrix was computed to assess how feature norm values across models relate to each other within each architecture family.

	ResNet18	**ResNet50**	**ResNet101**
ResNet18	1.000000	0.095631	0.095166
ResNet50	0.095631	1.000000	0.943153
ResNet101	0.095166	0.943153	1.000000

Observation:

- ResNet50 and ResNet101 show a strong correlation (**0.943153**), indicating similar feature norm behavior for deeper residual models.

- Correlation between ResNet18 and deeper variants is very weak (**~0.095**), suggesting that shallow and deep ResNet models respond differently at the feature norm level.

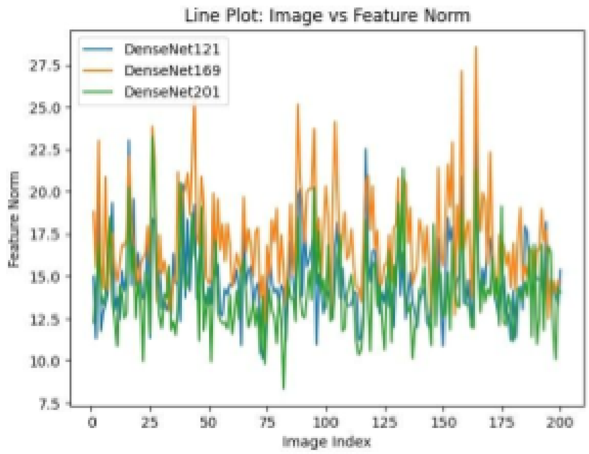
	DenseNet121	DenseNet169	DenseNet201
DenseNet121	1.000000	0.666858	0.523361
DenseNet169	0.666858	1.000000	0.501101
DenseNet201	0.523361	0.501101	1.000000

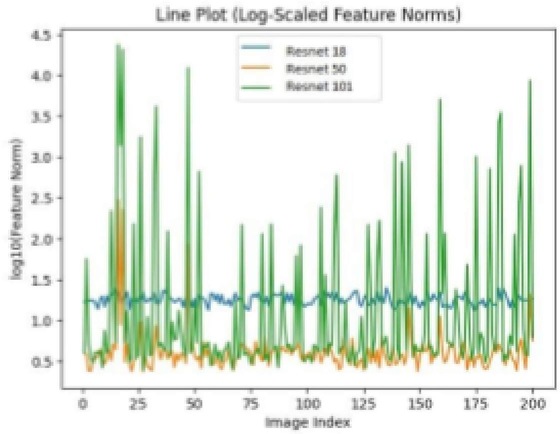
Observation:

- DenseNet models exhibit moderate positive correlations (**0.50–0.67**) across depths.
- Correlation strength decreases gradually with increasing depth difference, indicating consistent but not identical feature norm behavior within the DenseNet family.

Comparison: Correlation Patterns in ResNet vs DenseNet

- **Strength of Deep-Model Correlation:**
  Deeper ResNet models show very strong correlation (**ResNet50–ResNet101: 0.94**), whereas DenseNet deep models show only moderate correlation (**DenseNet169–DenseNet201: ~0.50**).
- **Shallow–Deep Relationship:**
  ResNet18 has very weak correlation with deeper ResNets (**~0.095**), while DenseNet121 maintains moderate correlation with deeper DenseNets (**0.52–0.67**).
- **Consistency Across Depths:**
  DenseNet correlations decrease gradually as depth increases, indicating smoother transitions across models. ResNet correlations show a sharp split between shallow and deep variants.
- **Overall Pattern:**
  Correlation structure within DenseNet is more uniform across depths, while ResNet exhibits strong alignment only among deeper models and weak alignment with shallow ones.





4. Per-Image Analysis

ResNet Family

Per-image feature norm analysis for ResNet models evaluates how individual images vary across ResNet18, ResNet50, and ResNet101 using metrics such as minimum, maximum, and range of feature norm values.

Observation:

- Certain images show low feature norms in ResNet18 but substantially higher norms in deeper variants, especially ResNet101.
- Several images exhibit very large per-image ranges due to extreme feature norm values in deeper ResNet models.
- Outlier images are primarily driven by ResNet50 and ResNet101, indicating depth-sensitive amplification for specific inputs.

DenseNet Family

Per-image feature norm analysis for DenseNet models examines variation across DenseNet121, DenseNet169, and DenseNet201.

Observation:

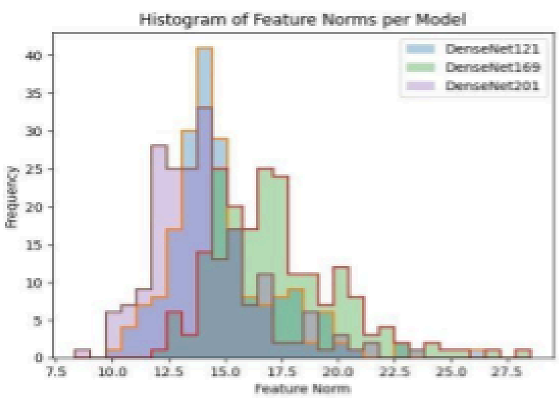
- Most images show consistent feature norm values across DenseNet depths, resulting in relatively small per-image ranges.
- Images that exhibit higher feature norms do so gradually across DenseNet variants rather than abruptly.

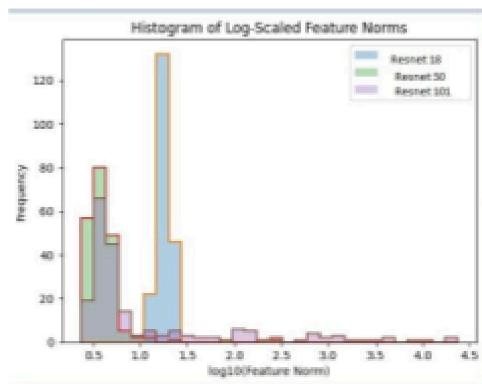
- Extreme per-image outliers are rare, indicating stable behavior across images.

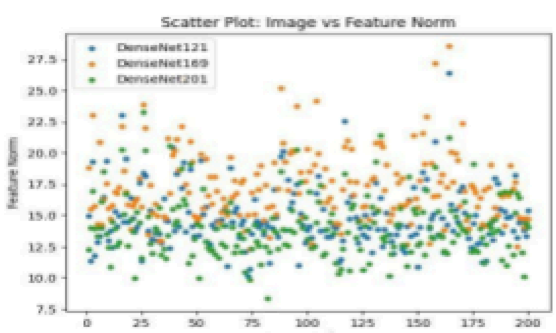
Comparison: Per-Image Behavior

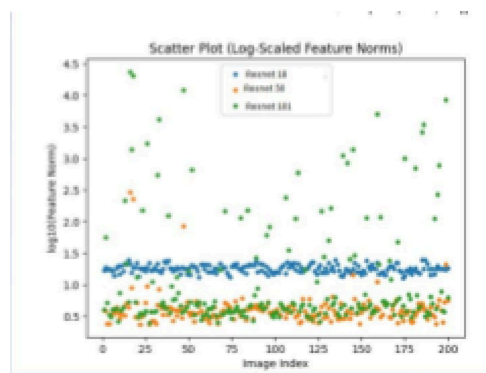
- **Range of Variation:**
  Per-image feature norm ranges are generally larger for ResNet models than for DenseNet models.

- **Outliers:**
  ResNet shows a higher frequency of images with extreme feature norm values, while DenseNet distributions remain compact.
- **Consistency:**
  DenseNet maintains more uniform per-image behavior across depths, whereas ResNet shows sharp per-image divergence between shallow and deep models.









5. Scaling Law Analysis

A power-law relationship was applied to evaluate how model depth affects feature norm magnitude:

$$\mathbf{F = k \cdot N^{(-\alpha)}}$$

where
F is the feature norm, **N**
is the model depth, **α** is
the scaling exponent.

Scaling exponents were estimated on a per-image basis and averaged to obtain an aggregate α value for each architecture family.

ResNet Family

For ResNet architectures (ResNet18, ResNet50, ResNet101), the average scaling exponent across all images is:

$$\boldsymbol{\alpha}(\textbf{ResNet}) = -\mathbf{0.326}$$

Observation:

- The negative α indicates that, on average, feature norms decrease with increasing depth in the ResNet family.
- This aggregate behavior is influenced by large variance and outlier feature norms in deeper ResNet models.
- Individual images may still exhibit positive or near-zero scaling, but the overall trend reflects unstable depth-dependent behavior.

DenseNet Family

For DenseNet architectures (DenseNet121, DenseNet169, DenseNet201), the average scaling exponent across all images is:

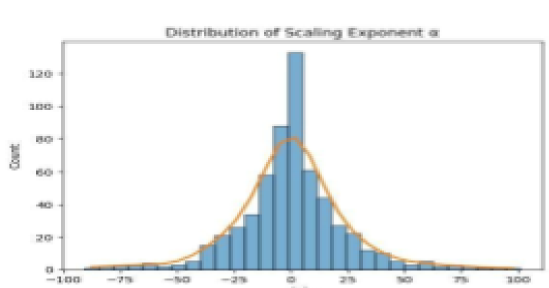
$$\boldsymbol{\alpha}(\textbf{DenseNet}) = -\mathbf{0.028}$$

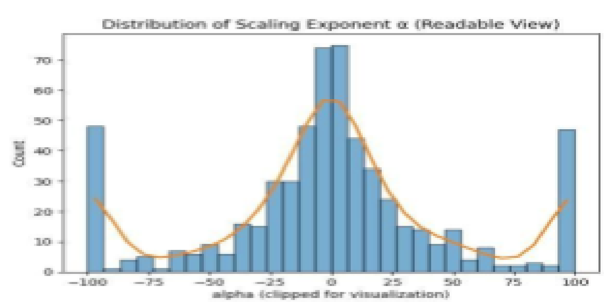
Observation:

- The α value is close to zero, indicating weak dependence of feature norm magnitude on depth.
- Feature norm scaling across DenseNet models remains stable and consistent across images.
- The near-flat scaling suggests controlled growth of representations with increasing depth.

Comparison: Scaling Behavior

- Magnitude of α:
  ResNet shows a strongly negative scaling exponent (α ≈ −0.326), while DenseNet exhibits near-neutral scaling (α ≈ −0.028).
- Depth Sensitivity:
  Feature norms in ResNet change substantially with depth, whereas DenseNet feature norms remain largely depth-invariant.
- Stability:
  DenseNet demonstrates stable scaling behavior across images, while ResNet scaling is dominated by variance and outliers.





Analysis of Feature Norm Scaling with Dataset Size Across ResNet and DenseNet Architectures

1. Introduction

This section presents a detailed quantitative analysis of feature norm values obtained from multiple convolutional neural network architectures evaluated across different dataset sizes (100, 200, and 300 samples), using two model families: ResNet (ResNet18) and DenseNet (DenseNet121), on unlabeled image data. The objective is to examine how dataset size influences feature magnitude, identify scaling patterns as the amount of training data increases, and analyze differences in scaling behavior between residual and densely connected network architectures.

2. Descriptive Statistics

Descriptive statistics were computed for both **DenseNet and ResNet** across different **dataset sizes (100, 200, and 300 images)** to summarize the distribution of feature norm values. Key statistics include mean, median, standard deviation, minimum, maximum, and quartiles.

DenseNet – Descriptive Statistics (Dataset Size Based)

Dataset Size	Mean Feature Norm	Median	Std Dev	Min	25%	75%	Max
100 Images	14.49	14.12	2.43	8.77	12.73	16.21	22.28
200 Images	15.12	14.66	2.40	10.60	13.77	15.90	27.30
300 Images	15.26	14.92	2.30	10.60	13.82	16.49	27.30

Observation:

- The dataset with **300 images** exhibits the highest mean and median feature norms for DenseNet, indicating stronger overall feature activation as dataset size increases.
- The **100-image and 200-image datasets** show comparable distributions, suggesting diminishing gains in feature magnitude beyond a certain dataset size.
- The relatively low and consistent standard deviation across all dataset sizes indicates stable and well-controlled feature representations in DenseNet.

ResNet – Descriptive Statistics (Dataset Size Based)

Descriptive statistics were computed for ResNet across increasing dataset sizes to summarize feature norm distributions.

Dataset Size	Mean Feature Norm	Median	Std Dev	Min	25%	75%	Max
100 Images	18.16	17.67	2.87	11.68	16.63	20.14	25.73
200 Images	18.15	17.84	2.67	12.94	16.07	19.83	25.20
300 Images	18.20	17.89	2.63	12.94	16.14	19.83	25.20

Observation:

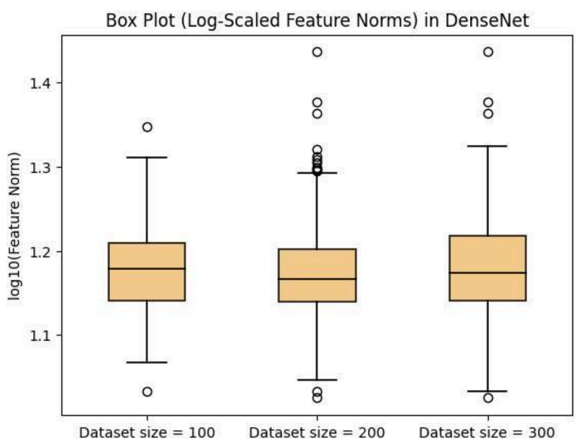
- Across all dataset sizes, ResNet shows **stable mean and median feature norms**, indicating consistent feature scaling as dataset size increases.
- The standard deviation slightly decreases from 100 to 300 images, suggesting improved stability in feature representations with larger datasets.
- The close alignment between mean and median values across all dataset sizes indicates minimal outlier influence and well-behaved feature norm distributions.

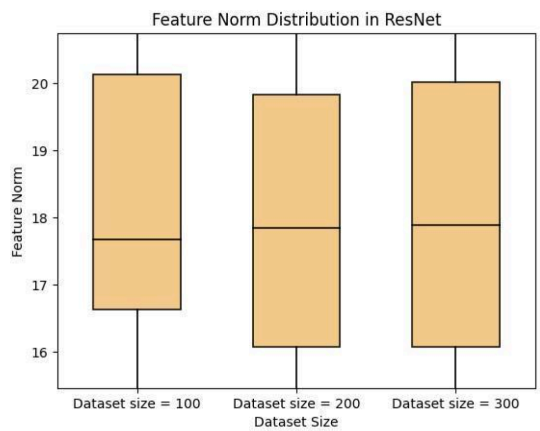
Comparison of Descriptive Statistics: ResNet vs DenseNet (Dataset Size Perspective)

- **Central Tendency:**
  DenseNet shows tightly clustered mean feature norms in the range **14.49–15.26**, with medians closely tracking the means across dataset sizes. ResNet exhibits higher mean feature norms, remaining stable around **18.15–18.20**, with medians also closely aligned.
- **Dispersion:**
  DenseNet maintains low and consistent standard deviations (**2.30–2.43**) across dataset sizes. ResNet shows slightly higher but stable dispersion (**2.63–2.87**), indicating controlled variability as dataset size increases.

- **Range and Extremes:**
  DenseNet maximum feature norms remain below **28** for all dataset sizes, while ResNet maximum values remain below **26**, indicating bounded feature activations for both models across datasets.

- **Quartile Spread:**
  Both architectures exhibit narrow and consistent interquartile ranges across dataset sizes, suggesting well-structured and symmetric feature norm distributions without extreme skewness.





3. Correlation Analysis

- A Pearson correlation matrix was computed to assess how **feature norm values across different dataset sizes (100, 200, and 300 images)** relate to each other within each architecture.

Dataset Size	**100 Images**	**200 Images**	**300 Images**
100 Images	1.000000	0.105141	0.083470
200 Images	0.105141	1.000000	0.428791
300 Images	0.083470	0.428791	1.000000

Observation:

- The **200-image and 300-image datasets** show a moderate positive correlation (**0.428791**), indicating more similar feature norm behavior as dataset size increases.
- Correlation between the **100-image dataset and larger datasets** is very weak (**0.08–0.10**), suggesting that feature norms learned from smaller datasets differ noticeably from those learned with more data.

Dataset Size	**100 Images**	**200 Images**	**300 Images**
100 Images	1.000000	0.036651	0.003931
200 Images	0.036651	1.000000	0.540270
300 Images	0.003931	0.540270	1.000000

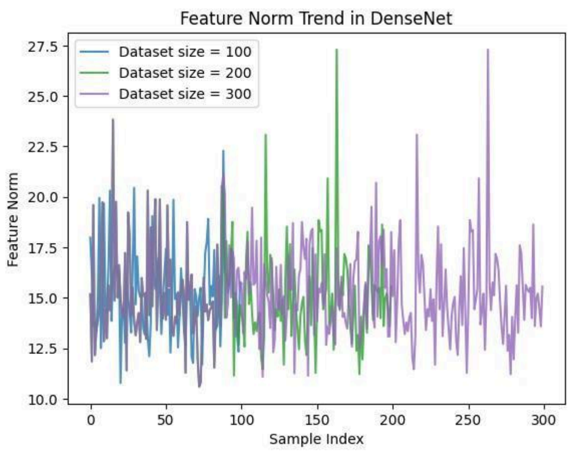
Observation:

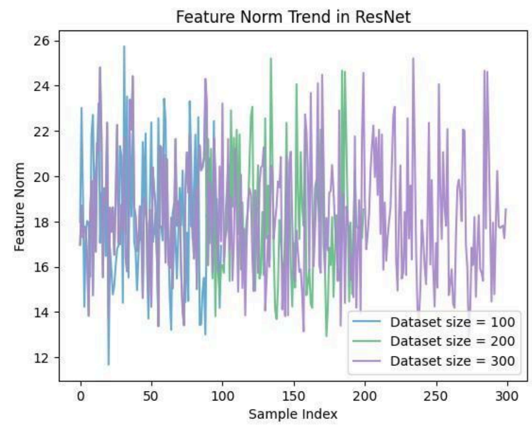
- DenseNet exhibits a **moderate positive correlation between 200 and 300 images** (**0.540270**), indicating consistent feature norm behavior at larger dataset sizes.
- Correlation involving the **100-image dataset is extremely weak** (≈0.00–0.04), highlighting significant differences in feature representations when trained on very limited data.

Comparison: Correlation Patterns in ResNet vs DenseNet (Dataset Size Perspective)

- **Strength of Large-Dataset Correlation:**
  Both architectures show stronger correlation between **200 and 300 images**, with DenseNet (**0.54**) exhibiting slightly higher alignment than ResNet (**0.43**).
- **Small vs Large Dataset Relationship:**
  Feature norms from the **100-image dataset** show weak correlation with larger datasets in both architectures, with DenseNet displaying a sharper drop than ResNet.
- **Consistency Across Dataset Sizes:**
  ResNet shows a gradual increase in correlation as dataset size grows, while DenseNet exhibits a clearer separation between small and large dataset regimes.

Overall Pattern:
Correlation structure suggests that **feature representations stabilize as dataset size increases**, with both architectures showing stronger alignment at higher data volumes and weak alignment when trained on limited data.





4. Per-Image Analysis

ResNet

Per-image feature norm analysis for **ResNet across different dataset sizes (100, 200, and 300 images)** evaluates how individual images vary as training data increases, using metrics such as minimum, maximum, and range of feature norm values.

Observation:

- Certain images exhibit lower feature norms when trained on **100 images** but show higher norms when trained on **200 and 300 images**, indicating data-dependent feature amplification.
- Several images demonstrate larger per-image ranges due to increased feature norm values at higher dataset sizes.
- Outlier images are more prominent in larger dataset settings, suggesting that increased data volume amplifies feature responses for specific inputs.

DenseNet

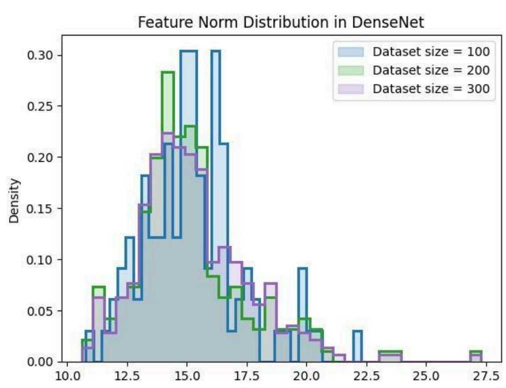
Per-image feature norm analysis for **DenseNet across dataset sizes (100, 200, and 300 images)** examines how individual images respond to increasing amounts of training data.

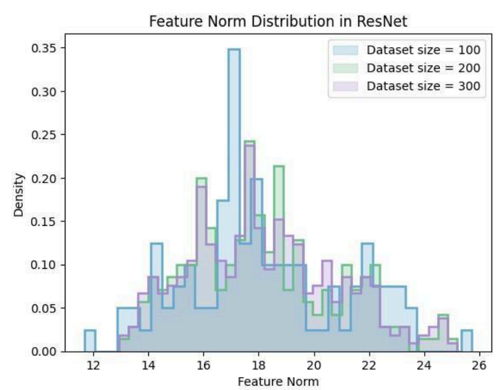
Observation:

- Most images show consistent feature norm values across dataset sizes, resulting in relatively small per-image ranges.
- Images that exhibit higher feature norms tend to increase **gradually** as dataset size grows rather than showing abrupt changes.
- Extreme per-image outliers are rare, indicating stable and controlled behavior across images even as dataset size increases.

Comparison: Per-Image Behavior (Dataset Size Perspective)

- **Range of Variation:**
  Per-image feature norm ranges are generally larger for **ResNet** than for **DenseNet** as dataset size increases.
- **Outliers:**
  ResNet shows a higher frequency of images with large feature norm values at higher dataset sizes, while DenseNet maintains compact per-image distributions.
- **Consistency:**
  DenseNet maintains more uniform per-image behavior across dataset sizes, whereas ResNet shows sharper per-image divergence as training data increases.





5. Scaling Law Analysis

A power-law relationship was applied to evaluate how **dataset size** affects feature norm magnitude:

$$\mathbf{F = k \cdot N\ \hat{}(-\alpha)}$$

where
F is the feature norm,

N is the dataset size (100, 200, 300 images),
α is the scaling exponent.

Scaling exponents were estimated on a **per-image basis** and averaged to obtain an aggregate **α** value for each architecture.

ResNet

For ResNet trained across different dataset sizes, the average scaling exponent across all images is:

$$\boldsymbol{\alpha}\ \textbf{(ResNet)} = -\ \mathbf{0.018}$$

Observation:

- The slightly negative α\alphaα indicates that, on average, feature norms show a **very weak decrease** as dataset size increases.
- This aggregate behavior reflects generally stable feature norm scaling with minor variation across images.
- While individual images may exhibit small positive or negative scaling trends, the overall relationship between dataset size and feature norm magnitude remains weak.

DenseNet

For DenseNet trained across different dataset sizes, the average scaling exponent across all images is:

$$\boldsymbol{\alpha}\ \textbf{(DenseNet)} \approx \mathbf{0.000}$$

Observation:

- The **α** value is effectively zero, indicating **negligible dependence** of feature norm magnitude on dataset size.

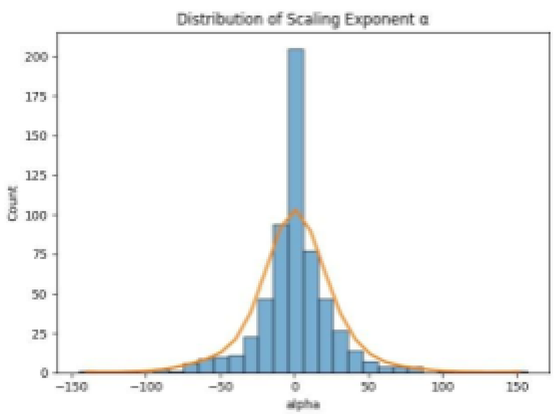
- Feature norm scaling remains highly stable and consistent across images as dataset size increases.

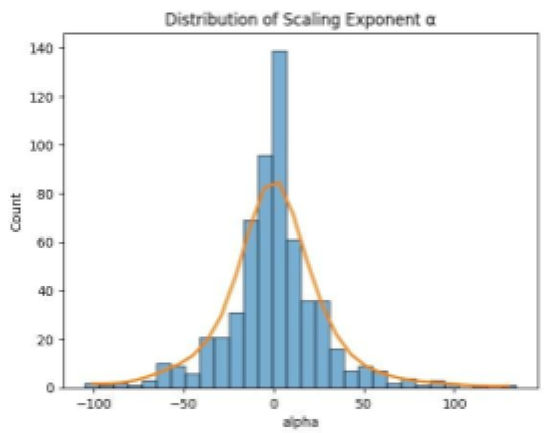
- The near-flat scaling behavior suggests controlled and dataset-invariant feature representations.

Comparison: Scaling Behavior (Dataset Size Perspective)

- **Magnitude of α:**
  ResNet shows a slightly negative scaling exponent (α ≈−0.018), while DenseNet exhibits near-neutral scaling (α≈0.000).
- **Dataset Sensitivity:**
  Feature norms in ResNet show marginal sensitivity to increasing dataset size, whereas DenseNet feature norms remain effectively dataset-size invariant.

- **Stability:**
  DenseNet demonstrates more stable scaling behavior across images, while ResNet exhibits minor variability without strong dataset-driven trends.





7. Conclusion

This work presented an empirical investigation of scaling behavior in convolutional vision models through the lens of internal representation norms, rather than task-level performance metrics. By analyzing ℓ2 feature norms across multiple architectures and dataset regimes under tightly controlled conditions, the study aimed to isolate how model capacity and data availability influence the structure and stability of learned representations.

Across model-size scaling experiments, clear architectural differences emerged. DenseNet variants exhibited stable, tightly clustered feature norm distributions with low variance and bounded extremes as depth increased. The near-zero scaling exponent observed for DenseNet indicates that feature magnitudes remain largely invariant to increasing model depth, suggesting effective internal normalization and controlled representation growth. In contrast, ResNet models showed highly depth-sensitive behavior, with deeper variants producing heavy-tailed distributions, extreme outliers, and rapidly increasing variance. The strongly negative scaling exponent estimated for the ResNet family highlights unstable depth-dependent scaling driven by a small subset of inputs with disproportionately large activations.

Dataset-size scaling experiments revealed a more subdued but consistent pattern. For both ResNet and DenseNet, increasing dataset size led to modest changes in average feature norms, with correlations strengthening as dataset size increased from 200 to 300 images. However, scaling exponents for dataset size remained close to zero for both architectures, indicating that representation magnitudes are largely invariant to dataset size within the examined range. This suggests that architectural design plays a more dominant role than data volume in shaping internal feature scale under frozen-backbone, controlled training conditions.

Per-image analyses further reinforced these findings. DenseNet demonstrated uniform and gradual per-image behavior across both model depth and dataset size, with rare extreme outliers and small per-image variation ranges. ResNet, by contrast, exhibited strong per-image divergence in deeper models and larger dataset regimes, indicating depth- and data-sensitive amplification effects for specific inputs. These differences highlight how architectural connectivity patterns influence not only aggregate statistics but also input-level stability.

Overall, the results indicate that **representation norm scaling in vision models is architecture-dependent, weakly influenced by dataset size, and strongly modulated by depth and connectivity design**. Dense connectivity promotes stable and depth-invariant representations, while residual architectures exhibit higher sensitivity to scaling, variance, and outliers as depth increases. These findings complement existing performance-based scaling studies by providing a representation-level perspective that is independent of supervised objectives.

This study demonstrates that internal representation norms offer a meaningful and interpretable signal for understanding scaling behavior in vision models. Future work can extend this analysis to additional architectures (e.g., Vision Transformers), broader dataset regimes, and alternative representation metrics to further elucidate the principles governing scalable and stable representation learning in computer vision.